COURSE : ZOO203

Current questions solved By:

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1) **Write Molecular methods advantages: imp**

i. they are useful with all organisms
ii. causes of the genetic variation and molecular processes are investigated that influence evolution
iii. provide information for the construction of evolutionary trees (phylogenies).

2) **Convergent Evolution vs divergent evolution: long**

the evolution of superficially similar structures in unrelated organisms is called convergent evolution, and the similar structures are said to be analogous.
Resemblance may also occur because two organisms share a common ancestry.

Divergent evolution:

it shows how species can change slightly over time and separate (diverge) into new forms.

3) **Directional selection:**

occurs when individuals at one phenotypic extreme are at a disadvantage compared to all other individuals in the population.

4) **Explain phylogenetic tree: (long)**

a branch represents an evolutionary connection between molecules (individuals, populations, or species).

The longer the branch, the greater the variation and the more distant the evolutionary relationship between molecules (individuals, populations, or species).
The relatively short branch distance between nodes reflects close evolutionary ties.
Modern Synthesis: the **fusion** (merger) of Mendelian **genetics** with Darwinian evolution that resulted in a unified theory of evolution. It is sometimes referred to as the **Neo-Darwinian theory**.

5) **Fossils are formed in sedimentary rock by a variety of methods**. Long imp

i. sediments cover an organism to prevent scavenging and seal out oxygen and slows decomposition. sediments continue to be piled on top of the dead organism, pressures build. Water infiltrates the remains and inorganic compounds and ions replace organic components. Hard parts of the organism are most likely to be fossilized, but delicate structures are sometimes fossilized when silica is involved with replacement. These pressure and chemical changes transform the organism into a stony replica.
ii. other fossilization processes form molds, casts of organisms, or carbon skeletons.
iii. tracks and burrows, and even mummified remains, are sometimes found.

6) **Define Adaptive traits:**

traits that promote successful reproduction are said to be adaptive.

Adaptivetraitsbecomemorecommoninsubsequentgenerations.Because organisms with maladaptive traits are less likely to reproduce, the maladaptive traits become less frequent a population.

7) **Reproductive isolation:**

when gene flow between populations or subpopulations does not occur.

When populations are reproductively isolated, natura lselection and genetic drift can result in evolution taking a different course in each subpopulation. Reproductive isolation can occur in different ways.

8) **Organic evolution and its type?** **long**

change in populations over time, or simply “descent with modification.”

The change must involve the genetic makeup of the population in order to be passed to future generations.

Microevolution: change in the frequency of alleles in populations over time is called microevolution.

Over longer timescales, microevolutionary processes result in large-scale changes.

Macroevolution: large-scale changes that result in extinction and the formation of new species are called macroevolution.

9) **Phyletic gradualism**

It is a model of evolution that depicts change as occurring gradually, over millions of years.

10) **Vestigial Structures:**

structures that have no apparent function in modern animals but clearly evolved from functioning structures in ancestors are called vestigial structures and provide another source of evidence of evolutionary change.

11) **Gene Flow and its impact: long**

changes in relative allelic frequency from the migration of individuals are called gene flow.

Effects of Gene Flow:

i. The exchange of alleles between an island population and a neighboring continental population, for example, can change the genetic makeup of those populations

ii. If gene flow continues and occurs in both directions, the two populations will become more similar.

iii. The absence of gene flow can make change in populations less likely.

12) **Lamarck: an early proponent of evolution: long**

Lamarckism (proposed by Jean Baptiste Lamarck (1744–1829)): this theory was based on a widely accepted theory of inheritance that organisms develop new organs, or modify existing organs, as needs arise.
Lamarck concluded that species could not become extinct—they simply evolved into different species.
Drawbacks of Lamarckism:

There is no evidence that changes in the environment can initiate changes in organisms that can be passed on to future generations. Instead, change originates in the process of gamete formation.

13) **Developmental Patterns ?** **long**

Evidence of evolution also comes from observing the developmental patterns of organisms.

The developmental stages of related animals often retain common features because changes in the genes that control the development of animals are usually harmful and are eliminated by natural selection.
Early embryonic stages of vertebrates are remarkably similar but adults in vertebrate groups are obviously different from one another. Differences in adult stages arise due to **evolution**.
These differences are a result of evolutionary changes in the genes that control the onset of developmental stages and the rate at which development occurs.
These changes result in differences in the size and proportions of organs.

14) **Explain Genetic Drift:** **long**

chance events influencing the frequencies of genes in populations result in genetic drift.
The chance of one or the other of two equally adaptive alleles being incorporated into a gamete, and eventually into an individual in a second generation, is equal.
Two special cases promote genetic drift:
i. when a subset of a larger population colonizes a new habitat
ii. when a population undergoes a near extinction event.

Genetic drift can cause big losses of genetic variation for small populations.

Founder effect: the loss of genetic variation that occurs when a new population is established by a very small number of individuals from a larger population.
Bottleneck effect: the loss of genetic variation when the number of individuals in a population is drastically reduced.

15) **Define Paleontology**: ?

the study of the fossil record, provides some of the most direct evidence for evolution.

16) **Define population?**

A group of individuals of the same species living and interbreeding within a given area.

17) **Define Vestigial Structure?**

Structures that have no apparent function in modern animals but clearly evolved from functioning structures in ancestors are called vestigial structures and provide another source of evidence of evolutionary change.

18) **Reproductive isolation**

When gene flow between populations or subpopulations does not occur.

19) **Disadvantages of small population.**

i. Small populations face a greater risk of extinction.

ii. Individuals in these populations can have a hard time finding quality mates so, fewer individuals mate and those that do risk inbreeding.
iii. Individuals in small population are more susceptible to random deaths.

20) **Disadvantages of large population.**

Carrying capacity: the maximum sustainable population size. Populations nearing their carrying capacity experience greater competition for resources, shifts in predator-prey relationships, and lowered fecundity. If the population grows too large, it may begin to exceed the carrying capacity of the environment and degrade available habitat.

21) **Development Pattern?**

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onset of developmental stages and the rate at which development occurs. These changes result in differences in the size and proportions of organs.

22) **Comparative anatomy?**

The study of the structure of living and fossilized animals and the homologies that indicate evolutionarily close relationships.

23) **Modes of Selection: ?**

Natural selection may affect a range of phenotypes in three ways.

i. **Directional selection**

occurs when individuals at one phenotypic extreme are at a disadvantage compared to all other individuals in the population.

ii. **Disruptive or diversifying selection:**

it involves circumstances selecting against individuals of an intermediate phenotype and produces distinct subpopulations.

iii. **Stabilizing selection:**

when both phenotypic extremes are deleterious, it narrows the phenotypic range and stabilizes the population mean.

24) **Drawbacks of Lamarckism**

There is no evidence that changes in the environment can initiate changes in organisms that can be passed on to future generations. Instead, change originates in the process of gamete formation.

25) **Speciation and its types? long**

Speciation

is the formation of new species. Speciation requires that subpopulations are prevented from interbreeding.

Types:

Allopatric speciation

Allopatric (Gr. allos, other 1 patria, fatherland) speciation occurs when subpopulations become geographically isolated from one another.

For example, a mountain range or river may permanently separate members of a population.

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Parapatric speciation

Another form of speciation, called parapatric (Gr. para, beside) speciation, occurs in small, local populations, called demes.

For example, all of the frogs in a particular pond or all of the sea urchins in a particular tide pool makee up a deme.

sympatric speciation

A third kind of speciation, called sympatric (Gr. sym, together) speciation, occurs within a single population. • Even though organisms are sympatric, they still may be reproductively isolated from one another.

26) **Neutral Mutation?**

Mutation that is neither harmful nor beneficial to organisms is neutral mutation. It may occur in regions of DNA that do not code for proteins

27) **Explain Adaptation**

occurs when a heritable change in a phenotype increases an animal's chances of successful reproduction in a specified environment.
Adaptations must be **heritable changes** to be passed to subsequent generations.

Adaptations are defined in the context of **enhancing reproductive success** because survival of a species occurs through successful reproduction, and survival of the species is the ultimate measure of success. Adaptations are defined in the context of a **specified environment** because a change that promotes successful reproduction in one environment may be detrimental to reproductive success in a different environment.
Adaptations may be:

- behavioral
- Physiological or iii. morphological.

29) Define Mutations:

changes in the structure of genes and chromosomes.

Mutations are the origin of all new alleles and a source of variation that may be adaptive for an animal.

Mutations make extinction less likely.

30) Effects of mutations:

The effects of mutations vary enormously.

Neutral Mutations:

neither harmful nor helpful to the organism. Neutral mutations may occur in regions of DNA that do not code for proteins.
Mutations that do affect protein function are more likely to be detrimental than beneficial. Mutations in DNA that is incorporated into a gamete have the potential to affect the function of every cell in an individual in the next generation. These mutations are likely to influence the evolution of a group of organisms.

Mutational equilibrium:

when a mutation from the wild-type allele to a mutant form is balanced by a mutation from the mutant back to the wild type.

Mutation pressure:

measure of the tendency for gene frequencies to change through mutation.

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